

UNIT-2

SIGNAL

- **Signal** is a way of communicating by sending information from one system to other system. Signal is a function that represents information or data (Gestures, actions, sounds, and expressions).
- Signal is an electromagnetic wave that carries information through physical medium. Here the data is converted into electromagnetic signal either as analog or digital and sent from sender to receiver.
- **There are two types of signal**
 - 1 analog signal
 - 2. digital signal

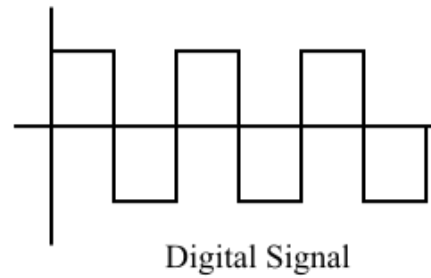
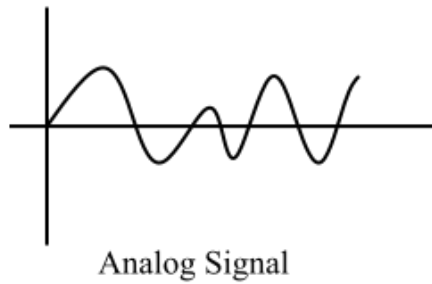
➤ **Analog signal**

- An analog signal is a continuous, time-varying signal that represents information as a continuous waveform.
- In an analog signal, the amplitude (magnitude) of the signal changes smoothly and continuously over time. This contrasts with a digital signal, which represents information using discrete values (typically 0s and 1s).
- Analog signals can take on a wide range of values within a certain range, and they are commonly used to represent real-world phenomena such as sound, light, temperature, voltage, and more. For example:
- **Audio Signals:** The varying air pressure caused by sound waves can be represented as analog signals. In older technology, music was stored on vinyl records, where the grooves on the record represented the analog waveform of the audio signal.

- **Electrical Signals:** In electronic circuits, voltage and current variations are often represented as analog signals. Devices like microphones and cameras produce analog signals that need to be converted to digital signals for processing in computers.
- **Sensor Data:** Many sensors, such as temperature sensors, pressure sensors, and light sensors, generate analog signals that need to be converted to digital form for processing and analysis.

➤ **Digital Signals**

- The signal, whose amplitude takes only limited values is called Digital signal.
- Digital signals are discrete, they contain only distinct values.
- Digital signals carry binary data i.e. 0 or 1 in form of bits, it can only contain one value at a period of time. Digital signals are represented as square wave or clock signals
- The minimum value is 0 volts whereas maximum value is 5 volts.
- Digital signals are less subjected to noise compared to analog signal.
- Transmission of digital data in analog channel is done by process called Modulation.
- **Examples of digital signals:**
 - Smart transmitters using various protocols transmit data through analog and digital signals.
 - Digital watches.
 - Digital video signals.
 - CD's.
 - DVD's.
 - Computer.



DIFFERENCE BETWEEN ANALOG AND DIGITAL SIGNAL

Parameter	Analog Signal	Digital Signal
Definition	A signal for conveying information which is a continuous function of time is known as analog signal.	A signal which is a discrete function of time, i.e. non-continuous signal, is known as digital signal.
Typical representation	An analog signal is typically represented by a sine wave function. There are many more representations for the analog signals also.	The typical representation of a signal is given by a square wave function.
Signal values	Analog signals use a continuous range of values to represent the data and information.	Digital signals use discrete values (or discontinuous values), i.e. discrete 0 and 1, to represent the data and information.

Parameter	Analog Signal	Digital Signal
Signal bandwidth	The bandwidth of an analog signal is low.	The bandwidth of a digital signal is relatively high.
Effect of electronic noise	Analog signals get affected by the electronic noise easily.	The digital signals are more stable and less susceptible to noise than the analog signals.
Accuracy	Due to more susceptibility to the noise, the accuracy of analog signals is less.	The digital signals have high accuracy because they are immune from the noise.
Power consumption	Analog signals use more power for data transmission.	Digital signals use less power than analog signals for conveying the same amount of information.

CHANNEL

- Communication systems, a channel refers to a pathway through which information (such as data, signals, or messages) is transmitted from a sender to a receiver. Channels can take various forms, including physical mediums like wires, optical fibers, and wireless frequencies, as well as abstract concepts in information theory.
- Here are some key points about channels in communication systems:

- **Physical Channels:** Physical channels are tangible mediums through which signals travel. Examples include electrical wires, coaxial cables, fiber optic cables, and radio waves in wireless communication. Each type of channel has its own characteristics and limitations, affecting the quality and speed of data transmission.
- **Bandwidth:** The bandwidth of a channel refers to the range of frequencies or data rates it can accommodate. A channel with higher bandwidth can transmit more data per unit of time.
- **Noise and Interference:** Channels can introduce noise and interference to the transmitted signals, which can degrade the quality of the received information. Engineers and designers work to minimize these effects through various techniques such as error correction codes and modulation schemes.
- **Baseband and Passband:** Baseband channels transmit signals directly over their original frequency range, while passband channels use modulation to shift the signal to a different frequency range for transmission. Passband channels are commonly used in wireless communication.
- **Multiplexing:** Channels can be multiplexed, meaning multiple signals can share the same physical medium. Time-division multiplexing (TDM) and frequency-division multiplexing (FDM) are common techniques.
- **Digital and Analog Channels:** Channels can transmit both digital and analog signals. Digital signals are discrete and composed of binary data, while analog signals are continuous and vary smoothly.

BIT-RATE:

- The Bit Rate is the number of bits that pass a given point in a telecommunication network in a given amount of time, usually a second.
- It is usually measured in some multiple of bits per second.
- For Example, Kilobits or thousands of bits per second (Kbps).
- The term bit rate is a synonym for data transfer rate.
- $\text{Bit Rate} = \text{Baud Rate} \times \text{Count of bits per signal unit.}$

MAXIMUM DATA RATE OF CHANNEL:

- It is the maximum amount of information that can be sent per second.
- It depends only on the bandwidth of the channel.
- There is an absolute upper limit to the data rate that can be achieved in any communication channel with a fixed bandwidth.

DATA RATE:

- The speed at which data is transferred within the computer or between a peripheral device.
- It is the speed at which data is transferred between two devices.
- It is measured in kilo bits per second or mega bits per second.

BANDWIDTH:

- It is the measurement of the ability of an electronic communications device or system to send and receive information.
- It is the capacity of a wired or wireless network communications link to transmit the maximum amount of data from one point to another over a computer network.
- Bandwidth connections can be symmetrical which means the data capacity is the same in both directions to upload or download data.

CHANNEL CAPACITY:

- It is the tight upper bound on the rate at which information can be reliably transmitted over a communication channel.
- It is a much-used metric for the maximum amount of traffic or signal that can move over a particular infrastructure channel.
- It is useful in computer science, in electrical engineering and in other disciplines evaluating the capacity of a channel.

BAUD RATE:

- It is the rate at which information is transferred in a communication channel.
- It is the count of signal units per second.
- It determines how many times the state of a signal is changing.
- It can determine how much bandwidth is required to send the signal.
- $\text{Baud Rate} = \text{Bit Rate} / \text{Number of bits per signal unit}.$

Difference between baseband and broad band transmission

Baseband Transmission	Broadband Transmission
It is a data transmission technique in which one signal needs the whole bandwidth of the channel to transfer the data.	It is a transmission technology in which many signals with different frequencies send data across a single channel at the same time.
It utilizes digital signals.	It utilizes analog signals.
The signals may be transmitted in both directions.	The signal may transmit only one direction.
It is bidirectional in nature.	It is unidirectional in nature.
It uses Time Division Multiplexing (TDM).	It uses Frequency Division Multiplexing (FDM).
It operates with bus topology.	It operates with both bus and tree topology.
It utilizes the same channel for sending and receiving data.	It utilizes two channels, one for transmission and the second for data reception.
Signals are only capable of travelling limited distances. Attenuation is needed for long distances.	Signals may be transmitted across long distances without attenuation.
It is simple and easy to install and maintain.	It is complex to install and maintain.
It is less expensive to design.	It is costly to design.
Manchester and differential Manchester encoding are used in baseband.	It doesn't utilize any digital encoding, but it utilizes the PSK (Phase shift keying) encoding.

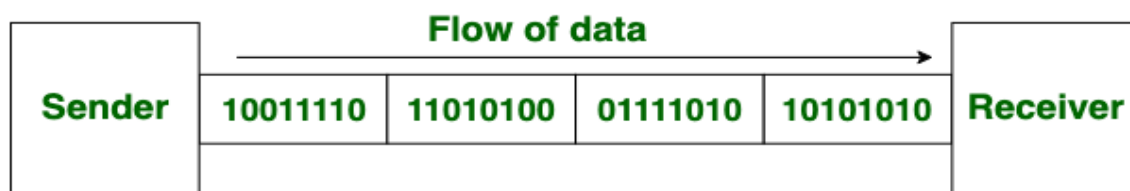
It contains a 50-ohm impedance.	It contains a 70-ohm impedance.
It utilizes coaxial cables, wires, and twisted-pair cables as the transfer medium for digital signals.	It sends digital signals via coaxial cable, optical fibre cables, and radio waves.
It is usually found in Ethernet.	It is usually found in telephone networks and cables.

SYNCHRONOUS TRANSMISSION:-

Synchronous Transmission: In Synchronous Transmission, data is sent in form of blocks or frames. This transmission is the full-duplex type. Between sender and receiver, synchronization is compulsory. In Synchronous transmission, There is no gap present between data. It is more efficient and more reliable than asynchronous transmission to transfer a large amount of data.

- **Example:**
- Chat Rooms
- Telephonic Conversations

Video Conferencing



Synchronous Transmission

ADVANTAGE

- It aids the user in transferring a huge amount of data.
- Every byte is sent without a pause before the next.
- It also helps to reduce timing errors.
- It allows connected devices to communicate in real-time.

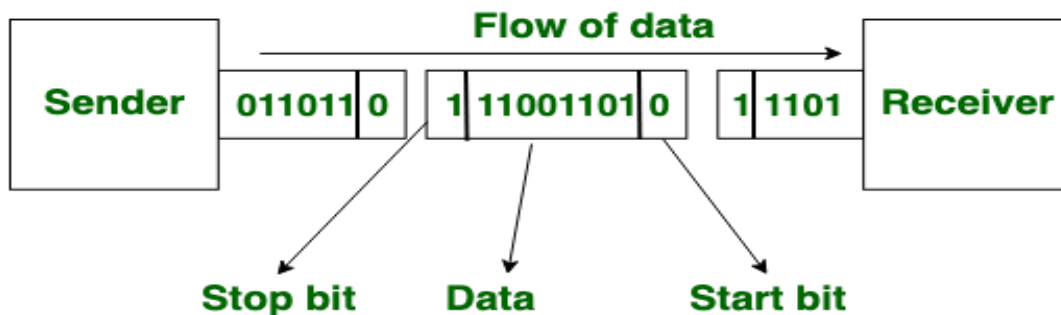
Disadvantages

- The sender and receiver must operate at the same clock frequency simultaneously.
- The accuracy of the received data is determined by the receiver's capacity to count the received bits precisely.

ASYNCHRONOUS TRANSMISSION

- **Asynchronous Transmission:** In Asynchronous Transmission, data is sent in form of byte or character. This transmission is the half-duplex type transmission. In this transmission start bits and stop bits are added with data. It does not require synchronization.

- Email
- Forums
- Letters



- **Asynchronous Transmission**

- **ADVANTAGE:**

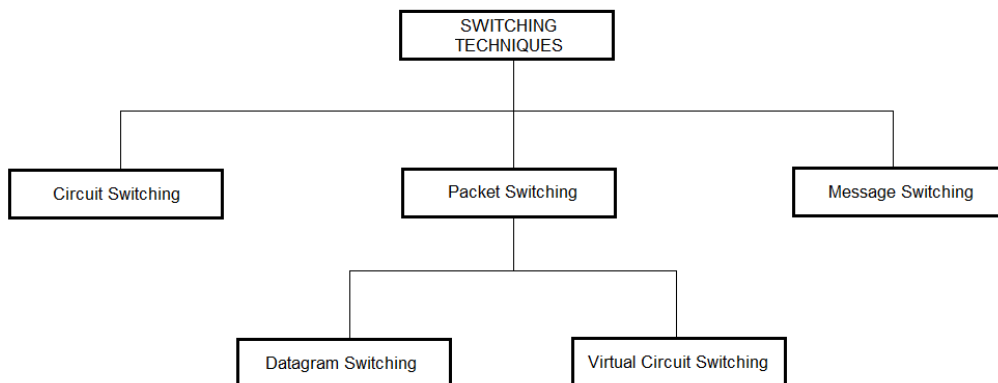
- It doesn't require synchronizing the receiver and transmitter.
- It is a very flexible technique of data transmission.
- This kind of transmission is simple to implement.
- It allows users to send signals from sources with varying bit rates.
- When the data byte transmission is complete, the data transmission may be resumed.

- **Disadvantages**

- The timing errors may occur because synchronization is difficult to determine.
- These bits could be mistakenly recognized due to the noise on the channel.
- The start and stop bits are extra bits that must be utilized in asynchronous transmission.
- It transmits information at a slower rate.

SWITCHING TECHNIQUES

- In large networks, there can be multiple paths from sender to receiver. The switching technique will decide the best route for data transmission.
- Switching technique is used to connect the systems for making one-to-one communication



Circuit Switching

- Circuit switching is a switching technique that establishes a dedicated path between sender and receiver.
- In the Circuit Switching Technique, once the connection is established then the dedicated path will remain to exist until the connection is terminated.
- Circuit switching in a network operates in a similar way as the telephone works.
- A complete end-to-end path must exist before the communication takes place.
- In case of circuit switching technique, when any user wants to send the data, voice, video, a request signal is sent to the receiver then the receiver sends back the acknowledgment to ensure the availability of the dedicated path. After receiving the acknowledgment, dedicated path transfers the data.
- Circuit switching is used in public telephone network. It is used for voice transmission.
- Fixed data can be transferred at a time in circuit switching technology.

Advantages of Circuit Switching:

- In the case of Circuit Switching technique, the communication channel is dedicated.
- It has fixed bandwidth.

Disadvantages of Circuit Switching:

- Once the dedicated path is established, the only delay occurs in the speed of data transmission.
- It takes a long time to establish a connection approx 10 seconds during which no data can be transmitted.
- It is more expensive than other switching techniques as a dedicated path is required for each connection.

PACKET SWITCHING

- The packet switching is a switching technique in which the message is sent in one go, but it is divided into smaller pieces, and they are sent individually.

- The message splits into smaller pieces known as packets and packets are given a unique number to identify their order at the receiving end.
- Every packet contains some information in its headers such as source address, destination address and sequence number.
- Packets will travel across the network, taking the shortest path as possible.
- All the packets are reassembled at the receiving end in correct order.

Datagram Packet switching:

- It is a packet switching technology in which packet is known as a datagram, is considered as an independent entity. Each packet contains the information about the destination and switch uses this information to forward the packet to the correct destination.
- The packets are reassembled at the receiving end in correct order.
- In Datagram Packet Switching technique, the path is not fixed.
- Intermediate nodes take the routing decisions to forward the packets.
- Datagram Packet Switching is also known as connectionless switching.

Virtual Circuit Switching

- Virtual Circuit Switching is also known as connection-oriented switching.
- In the case of Virtual circuit switching, a preplanned route is established before the messages are sent.

ADVANTAGES OF PACKET SWITCHING:

- It makes communications digital.
- Packet switching protocol has a high degree of error detection.
- The cost of packet switching is lower than that of circuit switching because the internal links between the switches are shared between many users.

DISADVANTAGES OF PACKET SWITCHING:

- Packet Switching protocols are complex.
- Data Transmission efficiency and Throughput is lower than that in circuit switched networks

MESSAGE SWITCHING

- Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.
- In Message Switching technique, there is no establishment of a dedicated path between the sender and receiver.
- The destination address is appended to the message. Message Switching provides a dynamic routing as the message is routed through the intermediate nodes based on the information available in the message.
- Message switches are programmed in such a way so that they can provide the most efficient routes.
- Each and every node stores the entire message and then forward it to the next node. This type of network is known as **store and forward network**

ADVANTAGES OF MESSAGE SWITCHING:

- It provides efficient traffic management.
- It provides efficient use of transmission channel.
- It reduces network traffic congestion.

DISADVANTAGES OF MESSAGE SWITCHING:

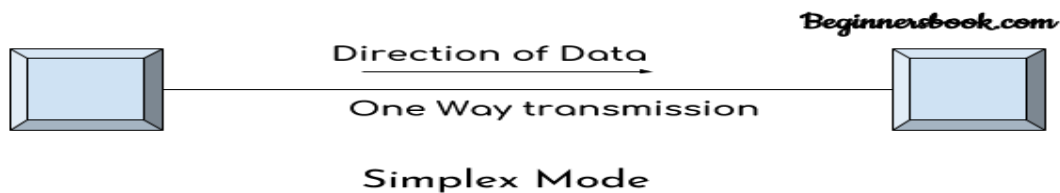
- It is not suitable for interactive applications.
- Each node requires large capacity for storing.

TRANSMISSION MODE:

- Transmission mode means transferring data between two devices. It is also known as a communication mode. Buses and networks are designed to allow communication to occur between individual devices that are interconnected.

1. **Simplex Transmission mode:-**

In Simplex mode, the communication is **unidirectional**, as on a one-way street. Only one of the two devices on a link can transmit, the other can only receive. The simplex mode can use the entire capacity of the channel to send data in one direction. Example: Keyboard and traditional monitors. The keyboard can only introduce input, the monitor can only give the output.



Advantages:

- Simplex mode is the easiest and most reliable mode of communication.
- It is the most cost-effective mode, as it only requires one communication channel.
- There is no need for coordination between the transmitting and receiving devices, which simplifies the communication process.
- Simplex mode is particularly useful in situations where feedback or response is not required,

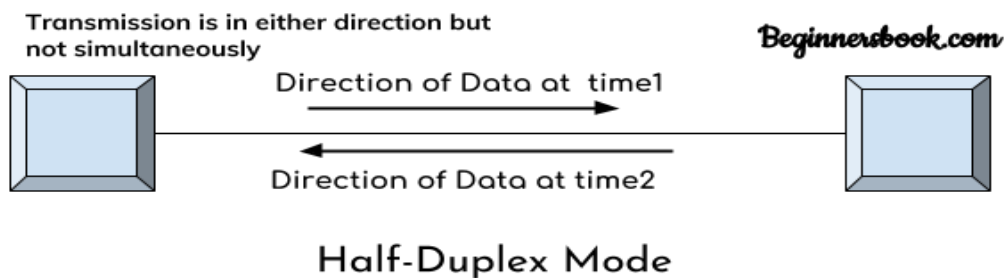
Disadvantages:

- Only one-way communication is possible.
- There is no way to verify if the transmitted data has been received correctly.
- Simplex mode is not suitable for applications that require bidirectional communication

2. Half-Duplex Mode –

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The half-duplex mode is used in cases where there is no need for communication in both directions at the same time.

Example: Walkie-talkie in which message is sent one at a time and messages are sent in both directions.



Advantages:

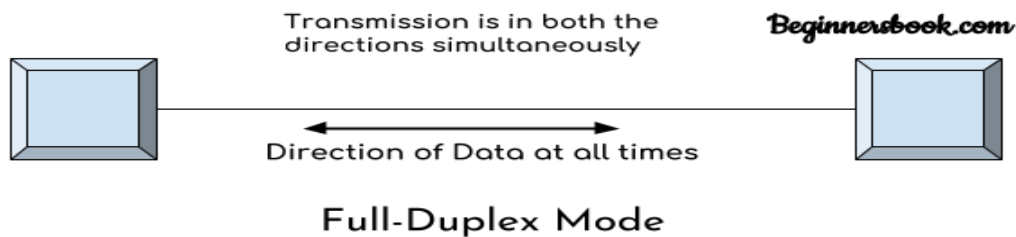
- Half-duplex mode allows for **bidirectional** communication, which is useful in situations where devices need to send and receive data.
- It is a more efficient mode of communication than simplex mode, as the channel can be used for both transmission and reception.
- Half-duplex mode is less expensive than full-duplex mode, as it only requires one communication channel.

Disadvantages:

- Half-duplex mode is less reliable than Full-Duplex mode, as both devices cannot transmit at the same time.
- There is a delay between transmission and reception, which can cause problems in some applications.
- There is a need for coordination between the transmitting and receiving devices, which can complicate the communication process.

3. Half-Duplex Mode –

- In full-duplex mode, both stations can transmit and receive simultaneously. In full duplex mode we can send the data and receive the data at same time.
- Full-duplex mode is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions. Example: Telephone Network in which there is communication between two persons by a telephone line, through which both can talk and listen at the same time



Advantages:

- Full-duplex mode allows for simultaneous bidirectional communication, which is ideal for real-time applications such as video conferencing or online gaming.
- It is the most efficient mode of communication, as both devices can transmit and receive data simultaneously.
- Full-duplex mode provides a high level of reliability and accuracy, as there is no need for error correction mechanisms.

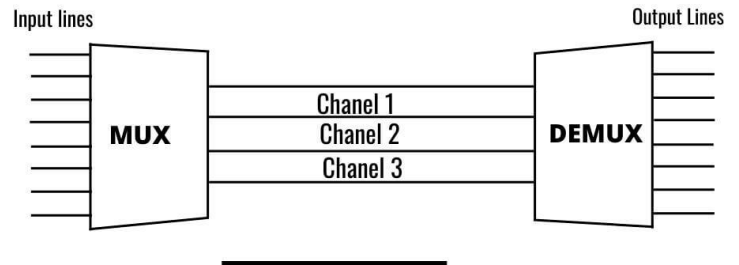
Disadvantages:

- Full-duplex mode is the most expensive mode, as it requires two communication channels.
- It is more complex than simplex and half-duplex modes, as it requires two physically separate transmission paths or a division of channel capacity.

- Full-duplex mode may not be suitable for all applications, as it requires a high level of bandwidth and may not be necessary for some types of communication

MULTIPLEXING

Multiplexing is a process in which multiple signals are combined together to create only one signal, which is then transmitted.

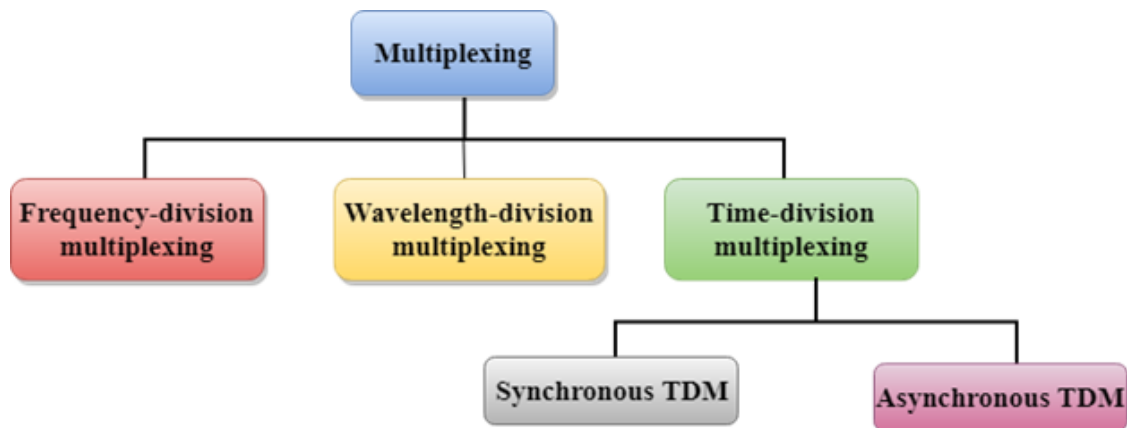


Multiplexing is the sharing of a medium or bandwidth.

It is the process in which multiple signals coming from multiple sources are combined and transmitted over a single communication/physical line.

Multiplexing is a technique through which many signals are transformed into just one signal. The hardware used to accomplish the process of multiplexing is called a Multiplexer, or simply a "MUX" for short.

TYPES OF MULTIPLEXING



1. FDM

- The full name of FDM is Frequency Division Multiplexing. It is the most commonly used technique in multiplexing.

- FDM is a technique in which signals of different frequencies are combined to create one signal.
- In FDM, signals are transmitted at the same time, but they have different frequencies.
- FDM is an analog technique where the transmission medium's bandwidth is divided into multiple channels.
- This technique is used in radio and television transmissions.

Advantages of FDM :

1. The FDM process is very simple.
2. It utilizes modulation.
3. It is used for analog signals.
4. It can transmit a large number of signals simultaneously.
5. It does not require any synchronization between the sender and receiver.
6. This technique is cost-effective.
7. It is reliable.

Disadvantages of FDM :

8. It requires a large number of modulators.
9. High bandwidth channels are needed for using FDM.
10. It requires more hardware compared to TDM (Time Division Multiplexing).
11. It is only used when there is a need for low-speed channels.
12. FDM systems can be quite expensive.

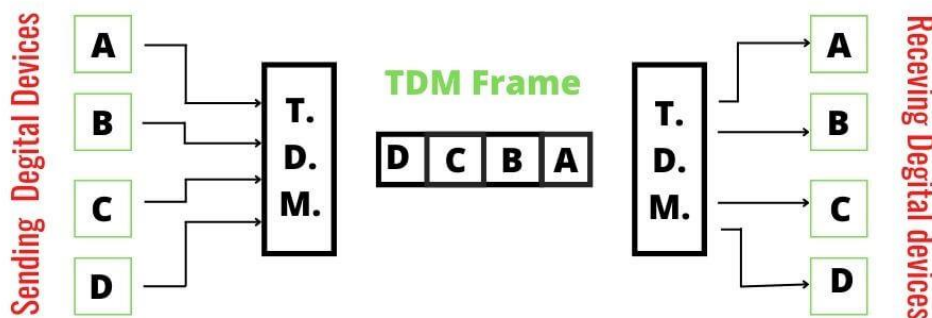
Applications of FDM

- FDM is used in radio and television broadcasting.
- It is employed in telephone systems.
- FDM is used for FM and AM broadcasting.
- It finds application in satellite communication systems.

- This technique assists in transferring data across multiple channels in satellite communication.
- It is used for broadband connections.
- FDM helps in transferring multimedia data such as video, audio, and images.
- In the past, FDM was used in cellular telephone systems.

2. TDM

- The full name of TDM is "Time Division Multiplexing".
- It is a type of digital technique used to combine digital signals into a single signal.
- In FDM (Frequency Division Multiplexing), signals are transmitted simultaneously in the same time but have different frequencies. However, in TDM, signals have the same frequency, but they are transmitted at different times.
- In this technique, data is not transferred together but is instead transferred sequentially one after the other. Rather than transferring bandwidth, TDM transfers time slots.



There are two types of TDM, which are explained in detail below.

1 - **Synchronous TDM** Synchronous TDM is a technique in which time slots are allocated to all devices in advance. Even devices that do not have any data to transmit are assigned time slots.

In Synchronous TDM, time slots are transmitted in the form of frames. If a device has no data to transmit, an empty slot is still transmitted.

2 - **Asynchronous TDM** Asynchronous TDM is a technique in which time slots are not allocated to all devices. Only those devices that have data to transmit are assigned time slots.

Asynchronous TDM is also known as Statistical TDM. It is considered a more efficient technique compared to synchronous TDM because it does not waste time slots like synchronous TDM.

Advantages of TDM (Time Division Multiplexing):

1. It is more flexible compared to FDM (Frequency Division Multiplexing).
2. Data can be transferred at a high speed in TDM.
3. It requires very little hardware.
4. TDM is less complex compared to FDM.
5. TDM is primarily used for digital signals, but it can also be used for analog signals.
6. It does not require any carrier wave or carrier signal.

Disadvantages of TDM:

1. It requires synchronization.
2. It requires address information and buffers.

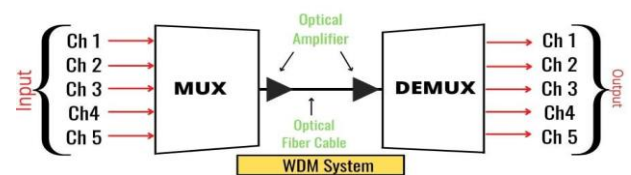
Applications of TDM:

1. TDM is used in digital audio mixing systems.
2. It is used in PCM (Pulse Code Modulation) transmission systems.
3. It is used in optical data transmission systems or optical fiber communications.
4. This technique is employed in SONET (Synchronous Optical Networking).

3. WAVELENGTH DIVISION MULTIPLEXING(WDM)

Wavelength Division Multiplexing (WDM) is a technique used to increase the capacity of optical fibers. It is an analog multiplexing technique that is similar to Frequency Division Multiplexing (FDM).

In WDM, optical signals are combined using a multiplexer to create a single large bandwidth of light.



Advantages of WDM:

1. WDM provides high bandwidth capacity.

- WDM offers a high bandwidth capacity.
- 2. It is reliable.
- WDM is reliable, ensuring stable and consistent data transmission.
- 3. It is a secure technology.
- WDM is considered a secure technology, helping protect data during transmission.
- 4. It is easy to implement.
- Implementing WDM is straightforward and user-friendly.
- 5. Reconfiguration is easy.
- Making changes or reconfiguring WDM systems is straightforward.

Disadvantages of WDM:

1. WDM involves optical components, which can increase the system's cost.
 - The inclusion of optical components in WDM systems can lead to higher costs.
2. It may not be highly scalable.
 - WDM systems may not be easily expandable to accommodate increasing demands.
3. Signals cannot be closely spaced in WDM.
 - WDM systems may not support very closely spaced signals due to limitations in wavelength spacing.

Applications of WDM:

- WDM is used in SONET networks.
- WDM is applied in SONET (Synchronous Optical Networking) networks.
- It is used in optical transport networks.
- WDM technology finds applications in optical transport networks.
- WDM is employed in local exchange networks.
- Local exchange networks utilize WDM technology for data transmission.

DATA ENCODING:

- It is the method by which certain communication devices encode digital data onto an analog signal for transmission.
- For example, telephone lines are analog but computer generates digital data, hence computers need to convert the digital data into analog data before it can be transmitted through telephone lines.

Different Data Encoding Techniques:

a. Digital Data Digital Signal:

- Digital Signal is a sequence of discrete, discontinuous voltage pulses.
- Each pulse is a signal element.
- Data is represented in binary.
- These binary data is transmitted by encoding each data bit into signal elements.

b. Digital Data Analog Signal:

- Digital Data is transmitted using analog signal through public telephone network.
- The telephone network can receive, switch and transmit analog signals in the voice frequency range of about 300 Hz to 3400 Hz.
- Modems are used for telephone network that produce signals in the voice frequency range.
- **For converting the digital data into analog signals, there types of encoding techniques are used:**
 - Amplitude Shift Keying (ASK)
 - Frequency Shift Keying (FSK)
 - Phase Shift Keying (PSK)

c. Analog Data Digital Signal:

- Analog data is converted into digital data and then it is suitable for transmission.
- The digital data can be transmitted.
- The digital data can be encoded as a digital signal using a code.
- **Codec uses two techniques as below:**
 - Pulse Code Modulation (PCM)
 - Pulse Amplitude Modulation (PAM)

d. Analog Data Analog Signal:

- Analog data in electrical form can be transmitted as base band signals easily and cheaply.
- Amplitude modulation, Frequency modulation and Pulse modulation techniques are used for transmitting analog signals.

MODULATION:

- Modulation is a process in which low-frequency signals are transformed into high-frequency signals.
- In other words, they are "modulated" because these low-frequency signals do not have enough energy to travel long distances effectively and tend to degrade quickly, making it difficult for them to reach their destination.
- Therefore, modulation is used to convert these low-energy signals into higher-frequency ones, allowing them to carry information over longer distances without significant loss and interference, ensuring they can reach the receiver effectively.

Need of modulation:-

- It is used for multiplexing
- Used for high bandwidth
- for the Long distance data transmission
- For increase the capacity of signals
- For reducing the noise. .

Modulation techniques used for Analog to Digital Conversion:

➤ **Pulse Amplitude Modulation (PAM):**

- It is the first step in converting an analog signal into digital signal.
- The technique takes an analog signal and samples it.

- The samples are taken at regular interval of time to generate the equivalent weighted signal.

- Sampling is the process of measuring the amplitude of the signal at equal intervals.

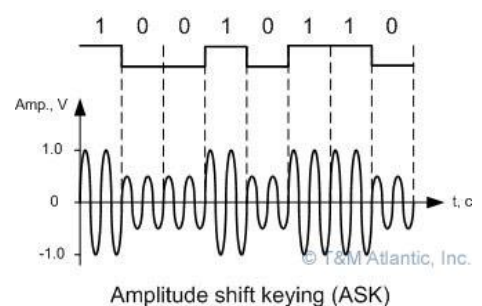
➤ **Pulse Code Modulation (PCM):**

- It is a very popular process in which analog signals are converted to digital form.
- It is a digital representation of an analog signal where the magnitude of the signal is sampled regularly at uniform intervals, then quantized to a series of symbols in a digital code.

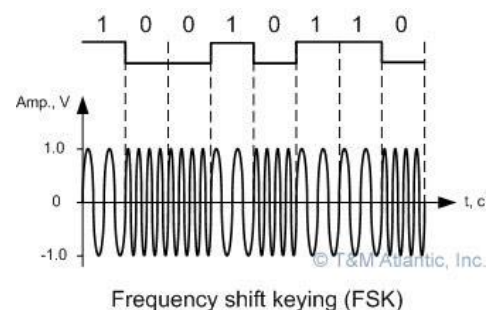
Modulation techniques used for Digital to Analog Conversion:

- Digital data is transmitted using analog signal through public telephone network.
- The telephone network can receive, switch and transmit analog signals.
- Digital to analog conversion is needed when the source of the signals is digital but the carrier of the signals is analog.
- There are three main methods to convert digital signals to analog signals as below:

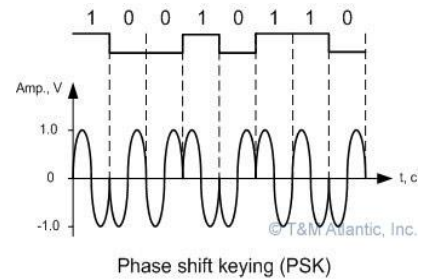
- **Amplitude Shift Keying (ASK):** ASK is the simplest form of digital modulation. In it, the strength of carrier signal is varied to represent binary 0 to 1. Here, both frequency and phase of the signal remain constant while the amplitude changes. One voltage represents binary 0 and other represents binary 1. This technique is normally used for transmitting data over optical fibers. The main disadvantage of Amplitude Shift Keying is that it is highly susceptible to noise interfaces.



- **Frequency Shift Keying (FSK):** In Frequency shift keying, the amplitude and phase remain constant but the frequency of the signal is varied to represent binary 0 or 1. One frequency represents binary 1 and the other frequency represents binary 0. The frequency of signal is constant during one bit and changes when the bit changes. FSK recovers the noise problem of ASK to a great extent, because the sender is only concerned about the frequency and not about the amplitude.



- **Phase Shift Keying (PSK):** In Phase Shift Keying, the amplitude and frequency remain constant but the phase of signal is changed to represent binary 0 or 1. The phase may be shifted at any two angles to represent binary 0 or binary 1.



TRANSMISSION MEDIA

Transmission media is the path through which information is conveyed from the sender to the receiver.

There are two types of transmission media:-

1. Wired transmission media(guided)
2. Wireless transmission media(unguided)

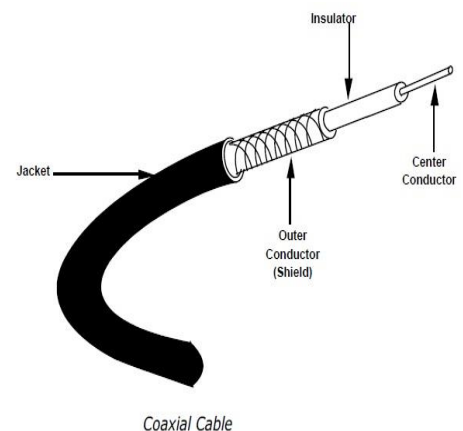
1. Wired transmissions media

- The transmission media in which the connection between two devices is established using physical methods such as cables or wires is referred to as "wired transmission media." It is also known as "guided media, **Bounded Media**."
- Point-to-point physical connection is needed.

TYPES OF TRANSMISSION MEDIA:-

I. COAXIAL CABLE:-

- Coaxial cable also known as "coax cable," is a type of wired transmission media commonly used for transmitting television signals, internet data, and other communication signals.
- It consists of a central conductor, an insulating layer, a metallic shield, and an outer insulating



layer.

- The central conductor carries the electrical signals, while the metallic shield surrounds it to prevent interference and signal loss.
- Coaxial cable is known for its ability to carry high-frequency signals over long distances with minimal signal degradation, making it suitable for various applications, including cable television, broadband internet, and some networking applications.

Advantages of Co-axial Cable:

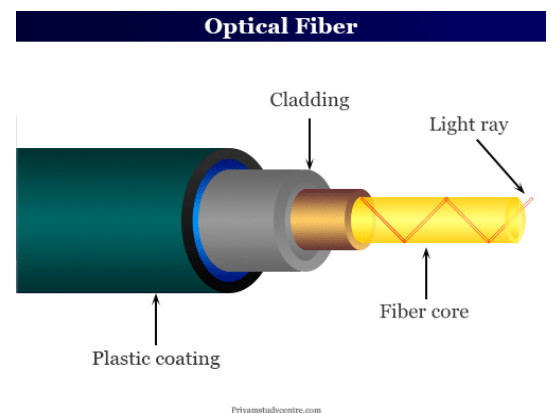
- It has better shielding than twisted pairs.
- It supports large number of channels per cable.
- It can span longer distances at higher speeds.
- It is used for both analog and digital data transmission.
- It has higher bandwidth.
- Relatively cheap as compared to fiber optic cables.

Disadvantages of Co-axial Cable:

- High cost as compared to twisted pair cables.
- Number of node connection is limited.
- High attenuation rate makes it expensive over long distances.
- Not compatible with twisted pair cables.

II. FIBER-OPTIC

- Fiber-optic cable is like a super-thin, flexible tube that carries information using light instead of electricity.
- Inside the cable, there is a tiny glass or plastic thread called an "optical fiber."
- When data or signals need to be sent, they're converted into beams of light that travel through these fibers.



- These beams of light can carry a lot of information really quickly, making fiber-optic cables perfect for transmitting data, like internet signals, over long distances with high speed and reliability.

Advantages of Fiber Optic Cables:

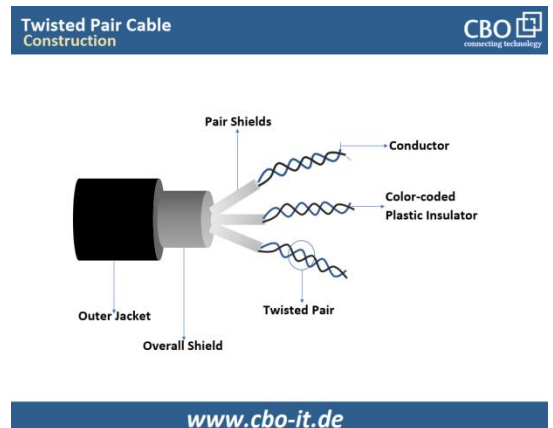
- It offers very high bandwidth than twisted pair and co-axial cables.
- Optical Fiber Cables are much lighter as compared to copper cables with the same bandwidth.
- It is highly suitable for environment where speed is needed with full accuracy.
- Data is transmitted digitally.
- These cables are ideally suited for broadband communication.
- They can carry various signals of different frequencies without mixing them.

Disadvantage of Fiber Optic Cables:

- It is more expensive.
- The Maintenance and Repairing of fiber optic cables is not only difficult but expensive also.
- Propagation of light is unidirectional.
- Thus, two fibers are needed for bidirectional communication.

III. Twisted pair

- It consists of two copper wires twisted together in a spiral fashion. The twisting helps reduce electromagnetic interference from external sources, which can degrade the quality of the signal.
- Twisted pair cables come in two main varieties: unshielded twisted pair (UTP) and shielded twisted pair (STP). UTP is commonly used for Ethernet networking, while STP has extra shielding to provide additional protection against interference.
- It is used for analog and digital signals.



- These are also commonly used within building for local area network.
- Twisted Pair Cables are used in telephone networks.
- Twisted Pair Cables are mainly used for low cost, low performance applications.

Advantages of Twisted Pair Cables:

- It is low cost medium that other guided media.
- It is the oldest and easily available transmission media.
- It is used for both analog and digital signals.
- It is flexible and easy to connect.
- It is best for short and medium range transmission.

Disadvantages of Twisted Pair Cables:

- These cables suffer from signal noise and low transmission rate.
- It has low bandwidth so these cannot carry multiple frequencies of broadband communications.

2. Wireless transmission media

- The transmission media in which can do communication without physical contact, meaning communication happens without wires, is called "wireless transmission media." It is also referred to as "unguided media, unbounded.
- The need of wireless transmission arises, when it is required to be online all the time.
- In this type of transmission, sending and receiving of signals are achieved by means of an antenna.

It includes the following types:

- (a) Radio waves
- (b) Microwaves
- (c) Infrared waves

1. RADIO WAVES

- Radio waves are mostly omnidirectional, meaning they spread out in all directions when transmitted from an antenna.
- These radio waves have a frequency range from 3 kHz to 1 GHz. They are relatively easy to establish and have high attenuation.

Applications of Radio Waves:

- These waves are normally used in voice communication.
- These are also used in Local Area Network in those buildings where wiring is difficult.
- These are useful for multicasting; radio, television etc

2. MICROWAVES:-

- The transmission of information in this medium is done through electromagnetic waves, which have wavelengths measured in centimeters. Their frequency is higher than 1 GHz, and they are produced by various types of transmitters.
- Microwave signals are used for transmitting data without use of cables.
- Microwave signals transmit data and voice over long distances without any wire or cable.
- It is a high-speed data transmission media
- Microwaves are used in various applications, including cooking food, mobile communication, and even in Wi-Fi technology.
- Microwaves are relatively inexpensive and require less space to setup antennas

3. INFRARED:-

- In the year 1800, Sir William Herschel was the first to discover and develop infrared radiation.
- This is a specific type of electromagnetic radiation known as infrared. Infrared waves are not visible to the human eye, but they can be felt as heat on the skin.

- Infrared waves find application in various technologies, including TV remote controls, wireless LAN (Local Area Network), CCTV (Closed-Circuit Television), missile guidance systems, and more.
- It is widely used for short range communication.
- The remote control used in TV, DVD etc. all use infrared communication.
- It is relatively directional, cheap and easy to build.

DIAL-UP CONNECTION:-

- Dial-up connection is an outdated method of connecting to the internet or other remote networks. It was widely used in the late 20th century and early 21st century before broadband and high-speed internet connections became prevalent. Here's how a dial-up connection worked:
- **Modem:** A dial-up connection requires a device called a modem (short for modulator-demodulator). Modems are used to convert digital data from your computer into analog signals that can be transmitted over a standard telephone line.
- **Telephone Line:** You need an active telephone line to establish a dial-up connection. The modem connects to your phone line using a standard telephone jack.
- **Dialing:** To initiate a connection, your computer instructs the modem to dial a phone number, just like you would when making a voice call. This number typically belongs to an Internet Service Provider (ISP) that provides dial-up access.
- **Data Transfer:** Data is transmitted in analog form over the phone line. This type of connection is relatively slow compared to modern broadband technologies, with typical speeds ranging from 28.8 Kbps to 56 Kbps (kilobits per second).
- **Internet Access:** Once the connection is established, you can use your computer to access the internet, send emails, browse websites, and perform other online activities, albeit at a much slower pace compared to high-speed connections.

Dial-up connections had several limitations and drawbacks:

- **Slow Speed:** Dial-up connections were notoriously slow, making it impractical for activities that require high bandwidth, such as streaming videos or downloading large files.
- **Tied Up Phone Line:** While using dial-up, your phone line was occupied, meaning you couldn't make or receive voice calls simultaneously.
- It can be used only for short distance data communication

DSL:

- DSL stands for “Digital Subscriber Line”.
- It is a medium for transferring data over regular phone lines and can be used to connect to the Internet.
- It is the latest modem technology that allows existing twisted pair telephone lines to transmit multimedia materials and high speed data.
- Transfer speed can range from hundreds of thousands of bits per second up to several million bits per second.
- DSL is also referred to as a broadband technology.
- DSL has become a reliable service to meet the needs of customers who demand high-speed Internet access.

Advantages of DSL:

- The speed of DSL is much higher than a regular modem.
- DSL service does not share the line with other network users in that area.
- Bad weather does not affect DSL service.
- One can leave the internet connection open and still use the telephone line for voice calls.
- DSL do not require new wiring.
- It can use the existing phone line.

Disadvantages of DSL:

- DSL service is not available in all areas.
- The connection is faster for receiving data than it is for sending data over the Internet.

Types of DSL:

High Bit Rate Digital Subscriber Line (HDSL):

- It is the first generation of DSL technology.
- HDSL service uses less bandwidth.
- HDSL speeds may range from 1.5 Mbps to 2.048 Mbps over distances up to 12000 feet from the central office.
- HDSL does not require repeaters, they may be used to up the distance to 36000 feet.

a) Very High Data Rate Digital Subscriber Line (VDSL):

- A second common type of DSL is very high data rate digital subscriber line (VDSL).
- It uses twisted pair copper telephone lines, with a range of speeds depending on actual line length.
- Also VDSL uses fiber-optic, twisted pair co-axial cable for short distances.

b) Asymmetric Digital Subscriber Line (ADSL):

- It is a method to use the existing analog local loop lines for digital data transfer.
- This technology achieves higher bandwidths on the existing local loops of twisted pair cables.
- ADSL technology is asymmetric because it supports different data rates for upstream and downstream data.
- It uses higher data transfer rates in downstream direction and lower data transfer rates in the upstream direction.

c) Rate Adaptive Asymmetric Digital Subscriber Line (RADSL):

- It is a variation of ADSL which automatically adjusts the connection speed depending on the quality and length of the telephone line.
- In RADSL, the broadband modem is configured at startup to test the phone line and adjust the data rate.
- RADSL typically operates at a lower data rate than regular ADSL.
- Like ADSL, RADSL provides relatively more bandwidth for downloads and less for uploads.
- RADSL needs twisted pair copper cable.

d) Symmetric Digital Subscriber Line (SDSL):

- SDSL is mainly used for the business users that use internet for downloading as well as uploading the data.

- It is called symmetric because it provides the same amount of bandwidth for both upstream and downstream.

COMMUNICATION SATELLITES / SATELLITE NETWORKS:

- The use of satellite networks for data / signal transmission is the major advancement in telecommunication technology.
- Communication satellites have facilitated the distances.
- This communication covers maximum area on earth for a particular transmitter.
- The satellite receives data / signals from one earth station, amplifies them and sends them to the other earth station.
- The earth station consists of satellite dish that functions as an antenna and communication equipment to transmit and receive data from satellites.

Advantages of Satellite Network:

- It is used for long distance communication.
- It is high speed data transmission.
- It can reach anywhere in the world.

Disadvantages of Satellite Network:

- It is very expensive.
- Risk of failure is high.

MODEM:

- It is an internal or external device used to transmit data over a phone or cable line through a computer.
- The word modem is short for modulator-demodulator.
- A phone or cable line is analog because they are continuous feeds of variable information.
- Computers are digital because they are only able to understand a code of ones and zeros.
- The analog waves are converted by the modem into a digital signal of ones and zeroes so that the computer can process the information.
- Modem is a shortened name for Modulator-Demodulator, which basically means that all data sent through it will be sent through telephone line (modulator) and on the other side, the demodulator part will transform sound back into data.

